

Research Summary

Six projects, and what each one actually found.

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danny-397.github.io · github.com/Danny-397

Six projects, built May–August 2026. Each entry states what the project does and what it actually measured — including the three where the central idea did not hold up.

QuantumSafe Scan · quantumsafescan.com — Finds cryptography a quantum computer would break, scores exposure 0–100, names the NIST-standard replacement. The hard part isn't matching `RSA`; it's *not* matching it in a comment or a dead code path. A hybrid AST-and-regex engine with reachability ranking took precision from **49.1% to 100%** on a 9-language benchmark at 100% recall over 50 seeded cases. Audited 10,938 files; of 37 PyPI packages, 86% had a finding. Exports SARIF and CycloneDX CBOM, so it drops into real CI.

RL-Trader · [4-page paper](#) — PPO from scratch, aimed at markets as a hostile testbed, asking whether an agent that looks brilliant once survives five runs. It doesn't. One seed shows **+275%** against a **+19%** benchmark; across five the same agent averages **−2.7%**, CI **[−31%, +27%]** — indistinguishable from buy-and-hold, and significantly *worse* on equities ($p \approx 0.002$). Its own significance tooling caught the false positive a naive version ships, reproducing Henderson et al. (2018) in a new domain. Adds an original surrogate-data test separating "the agent is weak" from "there is no signal."

AlphaZero-style Chess Engine · rlchess.xyz — Rules only: no openings, no human games, no evaluation function. Dual-headed ResNet, PUCT tree search, 4,672-move policy head, playable in-browser via ONNX Runtime Web. Scaling self-play 10× bought **14 Elo**. Diagnosed as a self-play *signal* problem, not a compute ceiling — 92% of games ended non-decisive, starving the value head of gradient. A classical alpha-beta control beat the random baseline 10–0, confirming the harness was sound.

AlphaGlyph · alphaglyph.org — Symbolic-regression alpha discovery, evaluated against its own overfitting. Combinatorial Purged Cross-Validation puts the **probability of backtest overfitting at 72%** across a 42-variant grid, with about **52%** of in-sample Sharpe surviving out-of-sample. The figure is window-dependent, and the report documents that instability rather than quoting the friendlier run. 169 tests in CI.

Tradeski · tradeski.dev — Live market, macro, and sentiment data in one dashboard: 11 ticker symbols, 7 FRED series hourly, a 29-symbol screener, a retrieval-grounded assistant, 75 tests. Its most useful lesson was a bug: live prices never reached the browser because the server emitted on one Socket.IO namespace while the client listened on another. Both halves were correct alone, and all 71 tests passed throughout — every one of them tested a pure function, so no test covered a contract *between* components. One now does.

Neural Canvas · github.com/Danny-397/neural-canvas — Painting by hand gesture through a webcam: 21 3-D landmarks per hand at ~60 inferences/second, fully on-device, one HTML file, no build step, no server. The hard problem isn't detection but estimating *intent* from a noisy 60 Hz signal — a fixed low-pass filter trades jitter against lag, so strokes use a one-euro adaptive filter whose cutoff rises with hand velocity.